Solutions to the Exercises of Chapter 6

Chaotic Attractors

Chaos: An Introduction to Dynamical Systems, Alligood–Sauer–Yorke

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0. Definitions and tools used

I recall the definitions from the chapter that the solutions rely on, and prove two small lemmas about periodic points of piecewise linear maps that are used in 6.2, 6.6 and 6.8.

- **Forward limit set** (Def. 6.1). $\omega(x_0) = \{x : \ \forall N, \varepsilon \ \exists n > N, \ |f^n(x_0) - x| < \varepsilon\}$.
- **Chaotic orbit** (Chapter 3). Not periodic, not asymptotically periodic, and with a positive Lyapunov exponent.
- **Chaotic set / attractor / chaotic attractor** (Def. 6.2). A chaotic set is $\omega(x_0)$ for a chaotic orbit with $x_0 \in \omega(x_0)$. An *attractor* is a forward limit set whose basin (the set of initial values attracted to it) has nonzero length/area/volume. A chaotic attractor is a chaotic set that is an attractor.
- **Neighborhood and fraction of time.** $N(r, S) = \{x : \operatorname{dist}(x, S) \le r\}$ and $F(x_0, S) = \lim_{n\to\infty} \frac{\#\{f^i(x_0) \in S:\ 1 \le i \le n\}}{n}$.
- **Natural measure** (Def. 6.14). $\mu_f(S) = \lim_{r\to 0} F(x_0, N(r, S))$ for closed S, provided all x_0 outside a set of measure zero give the same answer.
- **Measures.** (a) $\mu \ge 0$; (b) countable additivity on disjoint sets; (c) μ(whole space) $= 1$ (probability measure); (d) $\mu(f^{-1}(S)) = \mu(S)$ (invariance).
- **Piecewise expanding maps** (Sec. 6.3). f continuous on $I = [p_0, p_k]$, $|f'| \ge \alpha > 1$ except at finitely many points. $\{p_0 < \cdots < p_k\}$ is a *stretching partition* if each $f(A_i)$, $A_i = [p_{i-1}, p_i]$, is exactly a union of some A_j's. The *transition graph* has an arrow $A_i \to A_j$ iff $f(A_i) \supset A_j$. **Theorem 6.11(3):** if every ordered pair of symbols can be connected by an allowable path (the graph is strongly connected), then f has a dense chaotic orbit and I is a chaotic attractor.
- **Z-matrix recipe** (Sec. 6.6). For piecewise linear f with slope $\pm s_j$ on A_j, $Z_{ij} = 1/s_j$ if $A_j \to A_i$ and 0 otherwise. If $Zp = p$, $p \ge 0$, then the measure with density p_i on A_i is invariant, once normalized by $\sum_i p_i L_i = 1$ ($L_i = \text{length } A_i$). Equivalently

$$p_i = \sum_{j:\ A_j \to A_i} \frac{p_j}{s_j}. \tag{Z}$$

  If 1 is a simple eigenvalue and all other eigenvalues have modulus < 1, the measure is the natural measure.

Lemma A (closed paths give periodic points). Let f be as above with stretching partition $\{A_i\}$ and let $A_{i_0} \to A_{i_1} \to \cdots \to A_{i_n} = A_{i_0}$ be a closed path of length n in the transition graph. Then there is a point $x \in A_{i_0}$ with $f^n(x) = x$ and $f^j(x) \in A_{i_j}$ for $0 \le j < n$.

Proof. Put $J_n = A_{i_n}$ and, going backwards, $J_j = A_{i_j} \cap f^{-1}(J_{j+1})$. Since $f(A_{i_j}) \supset A_{i_{j+1}} \supset J_{j+1}$ and f is continuous and monotone on A_{i_j}, J_j is a closed interval mapped onto J_{j+1}. Thus $J := J_0 \subset A_{i_0}$ is a closed interval $[u, v]$ on which f^n is continuous and monotone with $f^n(J) = A_{i_0} = [l, r]$. If f^n is increasing on J: $f^n(u) = l \le u$ and $f^n(v) = r \ge v$; if decreasing: $f^n(u) = r \ge u$ and $f^n(v) = l \le v$. Either way $f^n(x) - x$ changes sign (or vanishes) on J, and the Intermediate Value Theorem gives a fixed point of f^n in J. □

Lemma B (exact period). In Lemma A, suppose $f^j(x)$ is not a partition point for any j, and the word $w = A_{i_0} A_{i_1} \cdots A_{i_{n-1}}$ is *primitive* (not a repetition of a shorter word). Then x has minimal period exactly n.

Proof. Each $f^j(x)$ lies in the *interior* of A_{i_j}, and interiors of different A_i are disjoint, so the symbol of $f^j(x)$ is determined. If $f^m(x) = x$ with $0 < m < n$, then $f^{m+j}(x) = f^j(x)$ for all j, so $A_{i_{m+j}} = A_{i_j}$ (indices mod n): the word w would be a repetition of a shorter word. Contradiction. □

Exercise 6.1

Show that a chaotic attractor cannot contain a sink.

Solution.

Setting. A *sink* is a periodic orbit $\{q, f(q), \ldots, f^{k-1}(q)\}$ (a fixed-point sink if $k = 1$) such that all eigenvalues of $Df^k(q)$ have modulus < 1 (in one dimension: $|(f^k)'(q)| < 1$). Put $g = f^k$, so q is a fixed-point sink of g.

Step 1: a trapping ball. Choose a norm on $\mathbb{R}^m$ in which $\|Dg(q)\| < 1$ (possible because the spectral radius is < 1; in dimension one just use $|\cdot|$). By continuity of Dg there are $\varepsilon > 0$ and $r < 1$ with $\|Dg(z)\| \le r$ for all z in the closed ball $B = \{z : \|z - q\| \le \varepsilon\}$. By the mean value inequality on the convex set B,

$$\|g(z) - q\| = \|g(z) - g(q)\| \le r\|z - q\| \qquad (z \in B).$$

Hence $g(B) \subset B$ and $g^m(z) \to q$ as $m \to \infty$ for every $z \in B$.

Step 2: if a sink point lies in a forward limit set, the orbit is asymptotically periodic. Suppose $q \in \omega(x_0)$. By definition of $\omega(x_0)$ there is n_0 with $f^{n_0}(x_0) \in B$. By Step 1, $f^{n_0+mk}(x_0) = g^m(f^{n_0}(x_0)) \to q$ as $m \to \infty$. Applying the continuous map f^i $(0 \le i < k)$: $f^{n_0+mk+i}(x_0) \to f^i(q)$. So the orbit of x_0 converges to the periodic orbit of q: it is *asymptotically periodic*, and $\omega(x_0) = \{q, \ldots, f^{k-1}(q)\}$.

Step 3: conclusion. A chaotic attractor is $A = \omega(x_0)$ for a chaotic orbit $\{f^n(x_0)\}$, and a chaotic orbit is by definition *not* asymptotically periodic. If A contained a point q of a sink, Step 2 would make the orbit of x_0 asymptotically periodic, a contradiction. Hence A contains no sink. □

Remarks. (i) The same argument shows that a forward limit set containing a sink is the sink orbit itself (Exercise T6.2). (ii) A chaotic attractor *can* contain repelling or saddle periodic points, e.g. the tent map's attractor $[0, 1]$ contains the repelling fixed point 0.

Exercise 6.2

For the piecewise linear maps of Figures 6.10(a), 6.13(a) and 6.14(a), find all periods for which there are periodic points.

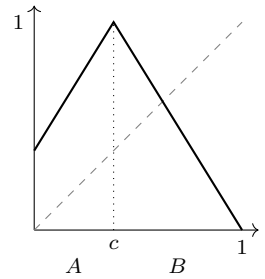


Fig. 6.10(a)

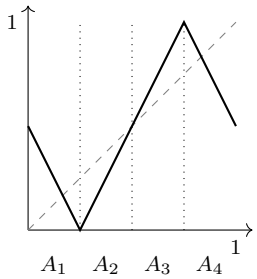


Fig. 6.13(a)

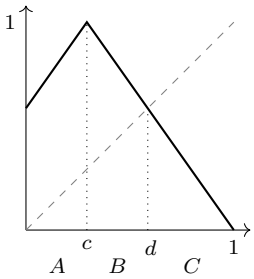
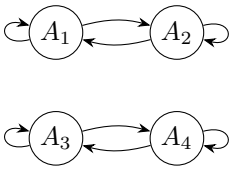
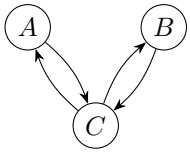


Fig. 6.14(a)







The three maps, their stretching partitions, and their transition graphs.

Solution. Write $a = (1+\sqrt{5})/2$ ($a^2 = a+1$). The three maps are (as in the text):

$$\text{Ex. 6.12 (Fig. 6.10a):}\quad f(x) = \begin{cases} \frac{1}{a+1} + ax, & 0 \le x \le c \\ a - ax, & c \le x \le 1 \end{cases}, \quad c = \tfrac{1}{a+1} \approx 0.382,$$

$$\text{Ex. 6.18 (Fig. 6.13a):}\quad f(x) = \begin{cases} \frac{1}{2} - 2x, & 0 \le x \le \frac{1}{4} \\ -\frac{1}{2} + 2x, & \frac{1}{4} \le x \le \frac{3}{4} \\ \frac{5}{2} - 2x, & \frac{3}{4} \le x \le 1 \end{cases},$$

$$\text{Ex. 6.19 (Fig. 6.14a):}\quad f(x) = \begin{cases} 2 + \sqrt{2}(x-1), & 0 \le x \le c \\ \sqrt{2}(1-x), & c \le x \le 1 \end{cases}, \quad c = \tfrac{2-\sqrt{2}}{2},\ d = 2 - \sqrt{2}.$$

In each case the plan is: read off the transition graph (a closed path of length n gives a point with $f^n(x) = x$ by Lemma A), check that the period is exactly n (Lemma B), and, when some periods are missing, prove that they really cannot occur.

(i) The map of Figure 6.10(a): all periods $n = 1, 2, 3, \ldots$

Here $f(A) = B$ and $f(B) = A \cup B$ for $A = [0, c]$, $B = [c, 1]$, so the graph is $A \to B$, $B \to A$, $B \to B$ (slopes $\pm a$, $a > 1$). The critical orbit is $c \to 1 \to 0 \to c$, so the partition points $\{0, c, 1\}$ form a period-3 orbit with symbols $0 \in A$, $c \in A \cap B$, $1 \in B$.

- $n = 1$: the loop $B \to B$ gives the fixed point $x = a/(a+1) = 1/a \approx 0.618$.
- $n = 2$: the loop $A \to B \to A$ gives the orbit $\{0.2764, 0.8292\}$ (explicitly $x = 1/(a+2)$ and $f(x)$).
- $n = 3$: the critical orbit $\{0, c, 1\} \approx \{0,\ 0.382,\ 1\}$.

- $n \geq 4$: use the closed path $A \to B \to B \to \cdots \to B \to A$, i.e. the word $w = AB^{n-1}$, which is primitive (it contains exactly one A). Lemma A gives x with $f^n(x) = x$. The only partition points that are periodic are $0, c, 1$, and along that orbit one never sees three consecutive B's ($0 \in A$, $c \in A \cap B$, $1 \in B$, then $0 \in A$), so for $n \geq 4$ the orbit of x avoids the partition points. Lemma B gives minimal period exactly n.

Answer: periodic points exist for every period $n \geq 1$. (The number of fixed points of f^n turns out to be the Lucas number $L_n = \mathrm{tr}\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)^n$: $1, 3, 4, 7, 11, \dots$)

(ii) The map of Figure 6.13(a): all periods $n = 1, 2, 3, \dots$

The partition $0 < \frac{1}{4} < \frac{1}{2} < \frac{3}{4} < 1$ gives $A_1, A_2 \to A_1, A_2$ and $A_3, A_4 \to A_3, A_4$: two disjoint copies of the "full 2-shift" graph. Better, each half is *conjugate to the tent map* $T(y) = 1 - |1 - 2y|$:

$$h_L(x) = 1 - 2x : \ [0, \tfrac{1}{2}] \to [0, 1], \qquad h_L \circ f = T \circ h_L \ \text{ on } [0, \tfrac{1}{2}];$$
$$h_R(x) = 2x - 1 : \ [\tfrac{1}{2}, 1] \to [0, 1], \qquad h_R \circ f = T \circ h_R \ \text{ on } [\tfrac{1}{2}, 1].$$

(Check on the left half: $f(x) = |2x - \frac{1}{2}|$ and $1 - 2|2x - \frac{1}{2}| = 1 - |4x - 1| = T(1 - 2x)$ since $1 - 2(1 - 2x) = 4x - 1$. On the right half: $f(x) = 1 - 2|x - \frac{3}{4}|$ and $2f(x) - 1 = 1 - |4x - 3| = T(2x - 1)$ since $1 - 2(2x - 1) = 3 - 4x$.)

The tent map has points of every period. T^n consists of 2^n linear laps of slope $\pm 2^n$, each mapping onto $[0, 1]$, so its graph meets the diagonal exactly once per lap: T^n has exactly 2^n fixed points. The fixed points of T^n include all points of period $m \mid n$, of which there are at most $\sum_{m|n,\, m<n} 2^m \leq 2^{\lfloor n/2 \rfloor + 1} < 2^n$ (for $n \geq 2$). So for each n there is a point of minimal period exactly n.

Explicit period-2 orbits of f (from the tent period-2 orbit $\{\frac{2}{5}, \frac{4}{5}\}$): $\{\frac{1}{10}, \frac{3}{10}\}$ in the left half and $\{\frac{7}{10}, \frac{9}{10}\}$ in the right half. Fixed points: $\frac{1}{4}, \ \frac{1}{2}, \ \frac{5}{6}$.

Answer: periodic points of every period $n \geq 1$ (in each of the two invariant halves).

(iii) The map of Figure 6.14(a): period 1 and all even periods; no odd period ≥ 3

The partition $0 < c < d < 1$ gives $A = (0, c)$, $B = (c, d)$, $C = (d, 1)$ with $f(A) = C$, $f(B) = C$, $f(C) = (0, d) = A \cup B \cup \{c\}$. The graph $A \to C$, $B \to C$, $C \to A$, $C \to B$ is *bipartite* between $\{A, B\}$ and $\{C\}$.

Which partition points are periodic? $c \mapsto 1 \mapsto 0 \mapsto d$, and $d = 2 - \sqrt{2}$ is a fixed point ($\sqrt{2}(1 - d) = d$). So $c, 1, 0$ are *eventually fixed*, not periodic, and d is the only periodic partition point (period 1).

No odd period ≥ 3. Let x be periodic and $x \neq d$. If some iterate of x equaled d, then since $f(d) = d$ the whole periodic orbit would be $\{d\}$, so no iterate is d. Likewise no iterate is 0, c or 1 (these land on d and stay there). So every iterate lies in $A \cup B \cup C$. Now $f(A) = f(B) = C$, and $f(C) \setminus \{c\} \subset A \cup B$ with c excluded. Hence the iterates alternate between C and $A \cup B$, and returning to x_0 forces the period to be *even*. Together with the fixed point d this shows: the only odd period that occurs is 1.

Every even period occurs. Period 2: the closed path $C \to A \to C$ gives the orbit $\{\frac{4-\sqrt{2}}{3}, \frac{2-\sqrt{2}}{3}\} \approx \{0.8619, 0.1953\}$. Check: $x = \frac{4-\sqrt{2}}{3} \in C$ has $1 - x = \frac{\sqrt{2}-1}{3}$, so $f(x) = \sqrt{2}(1 - x) = \frac{2-\sqrt{2}}{3} \approx 0.195 < c$,

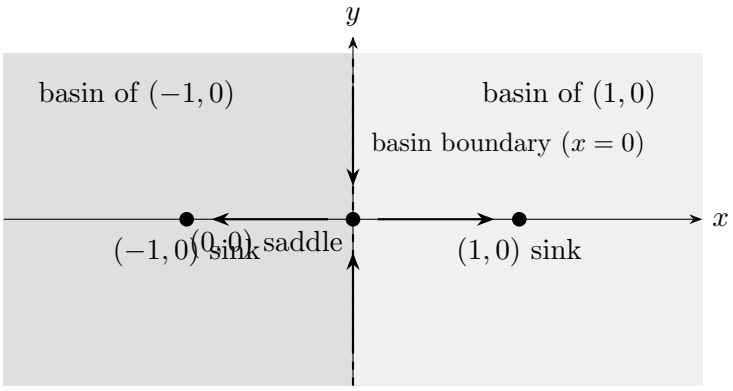
i.e. $f(x) \in A$; and then $f\big(\frac{2-\sqrt{2}}{3}\big) = (2-\sqrt{2}) + \sqrt{2}\cdot\frac{2-\sqrt{2}}{3} = \frac{(2-\sqrt{2})(3+\sqrt{2})}{3} = \frac{4-\sqrt{2}}{3} = x$. For $m = 2n$, $n \geq 2$, use the closed path $C\,A\,(C\,B)^{n-1}\,C$, whose word $w = CA(CB)^{n-1}$ is primitive (a single A). By Lemma A there is x with $f^{2n}(x) = x$ with this itinerary. The point x is not d because $f(x) \in \overline{A} = [0, c]$ while $d > c$; hence it is not on a periodic partition orbit, and Lemma B gives minimal period $2n$. (The path $C\,B\,C$ gives only the fixed point d.)

Answer: periods $1, 2, 4, 6, 8, \ldots$; no odd period ≥ 3.

Computer check. Solving $f^n(x) = x$ on every linear branch of f^n for $n \leq 10$ (resp. 9) reproduces exactly these period sets.

Exercise 6.3

Let $\mathbf{f}(x, y) = (\frac{4}{\pi}\arctan x,\ y/2)$. Find all forward limit sets, attractors and their basins. What is the basin boundary, and where do its points go?



Solution.

The x-coordinate. Let $g(x) = \frac{4}{\pi}\arctan x$. Then g is odd, strictly increasing, with $g'(x) = \frac{4}{\pi}\frac{1}{1+x^2}$, so $g'(0) = \frac{4}{\pi} \approx 1.27 > 1$ and $g'(\pm 1) = \frac{2}{\pi} \approx 0.64 < 1$.

Fixed points of g. $g(x) = x$ at $x = 0$ and at $x = \pm 1$ since $g(1) = \frac{4}{\pi}\cdot\frac{\pi}{4} = 1$. There are no others: let $h(x) = g(x) - x$. For $x > 0$, $h''(x) = -\frac{8}{\pi}\frac{x}{(1+x^2)^2} < 0$, so h is strictly concave on $[0, \infty)$ and can have at most two zeros there; these are 0 and 1. Consequently $h > 0$ on $(0, 1)$ and $h < 0$ on $(1, \infty)$. By oddness the fixed points are exactly $-1, 0, 1$; 0 is a source and ± 1 are sinks.

Orbits of g. If $0 < x < 1$: $h(x) > 0$ gives $g(x) > x$, and g increasing gives $g(x) < g(1) = 1$. So the orbit is increasing and bounded by 1, hence converges to a fixed point $\geq x > 0$, i.e. to 1. If $x > 1$: $h < 0$ gives $g(x) < x$, and $g(x) > g(1) = 1$, so the orbit decreases to 1. By oddness, $x < 0$ gives convergence to -1. If $x = 0$ the orbit stays at 0.

The y-coordinate. $y_n = y_0/2^n \to 0$ for every y_0.

Fixed points of f and their types. $Df = \operatorname{diag}\big(\frac{4}{\pi}\frac{1}{1+x^2}, \frac{1}{2}\big)$: at $(\pm 1, 0)$ the eigenvalues are $\frac{2}{\pi}, \frac{1}{2}$ (both < 1: sinks); at $(0, 0)$ they are $\frac{4}{\pi} > 1$ and $\frac{1}{2} < 1$ (a saddle).

Forward limit sets. For (x_0, y_0):

$$\omega(x_0, y_0) = \begin{cases} \{(1,0)\}, & x_0 > 0, \\ \{(-1,0)\}, & x_0 < 0, \\ \{(0,0)\}, & x_0 = 0 \quad (\text{the orbit } (0, y_0/2^n) \to (0,0)). \end{cases}$$

So the forward limit sets are the three points $(1,0), (-1,0), (0,0)$.

Attractors and basins. $\{(1,0)\}$ is an attractor with basin $\{x > 0\}$ (open right half-plane); $\{(-1,0)\}$ is an attractor with basin $\{x < 0\}$. The saddle $\{(0,0)\}$ is a forward limit set but **not** an attractor: it attracts only the y-axis, which has area zero.

Basin boundary. The boundary between the two basins is the y-axis $\{x = 0\}$. It is invariant: $\mathbf{f}(0, y) = (0, y/2)$. Its points move straight toward the origin along the axis and converge to the saddle $(0,0)$: the basin boundary is the *stable manifold* of the saddle. (The unstable manifold is the x-axis, which connects the saddle to the two sinks.)

Exercise 6.4

Let $f(x) = -x/2$. The attractor is $\{0\}$. Find $F(x_0, \{0\})$ and $F(x_0, N(r, \{0\}))$ for all x_0 and $r > 0$; compute $F(x_0, [0, \infty))$; find the natural measure μ_f.

Solution. The orbit is $x_n = (-\frac{1}{2})^n x_0$, so $x_n \to 0$, alternating in sign, and $x_n = 0$ iff $x_0 = 0$.

(1) $F(x_0, \{0\})$. If $x_0 = 0$, every iterate is 0: $F = 1$. If $x_0 \neq 0$, no iterate equals 0: $F = 0$.

$$F(x_0, \{0\}) = \begin{cases} 1, & x_0 = 0 \\ 0, & x_0 \neq 0. \end{cases}$$

(2) $F(x_0, N(r, \{0\}))$. Here $N(r, \{0\}) = [-r, r]$. Since $|x_n| = |x_0| 2^{-n} \leq r$ for all $n \geq n_0(r, x_0)$, all but finitely many iterates lie in $[-r, r]$, so the ratio in (6.5) tends to 1:

$$F(x_0, N(r, \{0\})) = 1 \qquad \text{for every } x_0 \text{ and every } r > 0.$$

(3) $F(x_0, [0, \infty))$. If $x_0 = 0$ all iterates are $0 \in [0, \infty)$, so $F = 1$. If $x_0 > 0$, then $x_n > 0$ exactly for even n; among $i = 1, \ldots, n$ there are $\lfloor n/2 \rfloor$ even ones, so the ratio is $\lfloor n/2 \rfloor / n \to \frac{1}{2}$. If $x_0 < 0$, $x_n > 0$ exactly for odd n, giving $\lceil n/2 \rceil / n \to \frac{1}{2}$.

$$F(x_0, [0, \infty)) = \begin{cases} 1, & x_0 = 0 \\ \frac{1}{2}, & x_0 \neq 0. \end{cases}$$

This is a "borderline" set (the attractor 0 lies on its boundary), and the answer depends on x_0, which is why the definition of natural measure uses $N(r, S)$.

(4) Natural measure. Let S be closed.

- If $0 \in S$ then $N(r, S) \supset [-r, r]$, so by (2) $F(x_0, N(r, S)) = 1$ for all $r > 0$, all x_0; hence $\mu_f(S) = 1$.
- If $0 \notin S$ then $\delta := \operatorname{dist}(0, S) > 0$. For $r < \delta/2$ the set $N(r, S)$ is disjoint from $[-\delta/2, \delta/2]$, and the orbit eventually stays in $[-\delta/2, \delta/2]$; so $F(x_0, N(r, S)) = 0$, and $\mu_f(S) = 0$.

The answer is the same for every x_0 (no exceptional set needed). Therefore

$$\mu_f(S) = \begin{cases} 1, & 0 \in S \\ 0, & 0 \notin S \end{cases}$$

i.e. μ_f is the **atomic measure at** 0 (see 6.9). It is also invariant: $0 \in f^{-1}(S) \iff f(0) = 0 \in S$.

Exercise 6.5

Let $g(x) = 2x(1-x)$ on $[0,1]$, which has a source at 0 and a sink at $1/2$. What natural measure does g generate?

Solution.

An exact formula for the orbits. Completing the square, $g(x) = \frac{1}{2} - 2(x - \frac{1}{2})^2$, i.e. $1 - 2g(x) = (1-2x)^2$. Iterating,

$$1 - 2x_n = (1-2x_0)^{2^n}, \qquad \text{so} \qquad x_n = \tfrac{1}{2}\Big[1 - (1-2x_0)^{2^n}\Big] \quad (n \geq 1).$$

(Check: $n = 1$ gives $\frac{1}{2}[1 - (1 - 4x_0 + 4x_0^2)] = 2x_0(1-x_0)$.)

Where orbits go. If $0 < x_0 < 1$ then $|1 - 2x_0| < 1$, hence $(1-2x_0)^{2^n} \to 0$ and $x_n \to \frac{1}{2}$ (superexponentially fast, since $g'(\frac{1}{2}) = 0$). If $x_0 = 0$ the orbit is the fixed point 0; if $x_0 = 1$ then $x_1 = 0$, and the orbit stays at 0. So *every* orbit in $[0,1]$ except those starting at 0 and 1 converges to the sink $\frac{1}{2}$. The exceptional set $\{0,1\}$ has length zero.

Natural measure. As in Exercise 6.4: for a closed $S \subset [0,1]$ with $\frac{1}{2} \in S$, $N(r,S) \supset [\frac{1}{2} - r, \frac{1}{2} + r]$, which eventually contains the whole orbit, so $F(x_0, N(r,S)) = 1$ for all $x_0 \in (0,1)$; if $\frac{1}{2} \notin S$ the orbit eventually leaves a neighborhood of S for good and $F = 0$ for small r. Therefore

$$\mu_g(S) = \begin{cases} 1, & \frac{1}{2} \in S \\ 0, & \frac{1}{2} \notin S \end{cases}$$

the **atomic measure at the sink** $x = \frac{1}{2}$. The source at 0 does not contribute: the only initial conditions that generate a different measure (the atomic measure at 0) are $x_0 \in \{0,1\}$, a set of measure zero, which Definition 6.14 allows us to ignore.

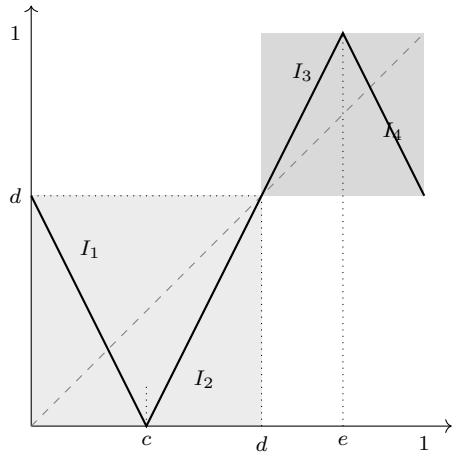
Exercise 6.6

Let f be the map of Example 6.19 and $g = f^2$. Sketch g, explain why it has no natural measure, and find its invariant measures.

Solution. Recall $c = \frac{2-\sqrt{2}}{2} \approx 0.293$, $d = 2 - \sqrt{2} \approx 0.586$, $f(x) = d + \sqrt{2}x$ on $[0,c]$ and $f(x) = \sqrt{2}(1-x)$ on $[c,1]$, with $f(0) = d$, $f(c) = 1$, $f(d) = d$, $f(1) = 0$.

Computing $g = f \circ f$. The extra break point of g is where $f(x) = c$: only the second branch can give this, $\sqrt{2}(1-x) = c$, i.e. $e = 1 - \frac{c}{\sqrt{2}} = \frac{3-\sqrt{2}}{2} \approx 0.793$.

- $0 \leq x \leq c$: $f(x) \in [d,1] \subset [c,1]$, so $g(x) = \sqrt{2}\,(1 - d - \sqrt{2}x) = 2 - \sqrt{2} - 2x = d - 2x$.
- $c \leq x \leq e$: $f(x) = \sqrt{2}(1-x) \in [c,1]$, so $g(x) = \sqrt{2}\big(1 - \sqrt{2}(1-x)\big) = 2x - (2 - \sqrt{2}) = 2x - d$.



The map $g = f^2$ of Exercise 6.6. It maps $[0, d]$ onto itself and $[d, 1]$ onto itself (shaded squares), with slopes ± 2.

- $e \le x \le 1$: $f(x) \in [0, c]$, so $g(x) = d + \sqrt{2} \cdot \sqrt{2}(1 - x) = d + 2 - 2x = (4 - \sqrt{2}) - 2x$.

Values at the break points: $g(0) = d$, $g(c) = 0$, $g(e) = 1$, $g(1) = d$; slopes $-2, +2, -2$. Also $g(d) = d$.

Two invariant intervals. g maps $[0, d]$ *onto* $[0, d]$ (values from $g(0) = d$ down to $g(c) = 0$ and back up to $g(d) = d$) and maps $[d, 1]$ *onto* $[d, 1]$ (from $g(d) = d$ up to $g(e) = 1$ and down to $g(1) = d$). Equivalently: f swaps $[0, d]$ and $[d, 1]$ ($f([0, d]) = [d, 1]$, $f([d, 1]) = [0, d]$), so f^2 preserves each.

Why there is no natural measure. Take the closed set $S = [0, c]$. If $x_0 \in [0, d)$ the whole orbit stays in $[0, d]$, where g has the 2×2 Z-matrix $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ (eigenvalues $1, 0$), so for almost every such x_0 the orbit is distributed according to normalized length μ_1 on $[0, d]$, and $F(x_0, N(r, S)) \to \mu_1(S) = c/d = \frac{1}{2} > 0$ as $r \to 0$. If $x_0 \in (d, 1]$ the orbit stays in $[d, 1]$ forever, never within $r < d - c$ of S, so $F(x_0, N(r, S)) = 0$. These two sets of initial values both have positive length, and they give different answers, so Definition 6.14 fails: g **has no natural measure**. (For f itself the orbit alternates between the two intervals, so every typical orbit sees both and a natural measure exists—it is the average of the two measures below.)

Invariant measures. Take the partition $0 < c < d < e < 1$: $I_1 = [0, c]$, $I_2 = [c, d]$, $I_3 = [d, e]$, $I_4 = [e, 1]$. Then $g(I_1) = g(I_2) = [0, d] = I_1 \cup I_2$ and $g(I_3) = g(I_4) = [d, 1] = I_3 \cup I_4$, all slopes ± 2. The Z-matrix is

$$Z = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

with eigenvalues $1, 1, 0, 0$. The eigenspace for 1 is two-dimensional, spanned by $(1, 1, 0, 0)$ and $(0, 0, 1, 1)$. Hence the invariant measures with piecewise constant density are

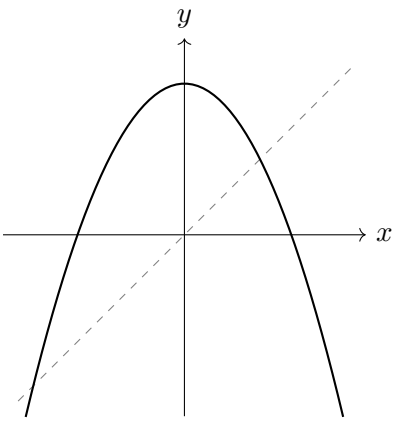
$$\boxed{\mu_\lambda = \lambda\,\mu_1 + (1 - \lambda)\,\mu_2, \quad 0 \le \lambda \le 1,} \qquad p(x) = \begin{cases} \dfrac{\lambda}{d}, & 0 \le x \le d \\ \dfrac{1 - \lambda}{1 - d}, & d \le x \le 1 \end{cases}$$

where μ_1, μ_2 are normalized length on $[0,d]$ and on $[d,1]$, $d = 2-\sqrt{2}$, $1-d = \sqrt{2}-1$. (Normalization: $p_1(c+c) + p_3(\ell_3+\ell_4) = \lambda + (1-\lambda) = 1$.) Each of μ_1, μ_2 is the natural measure of g restricted to its interval. The case $\lambda = \frac{1}{2}$ is the natural measure of f in (6.13): densities $\frac{1}{2d}$ and $\frac{1}{2(\sqrt{2}-1)} = \frac{1}{\sqrt{2}\,d}$.

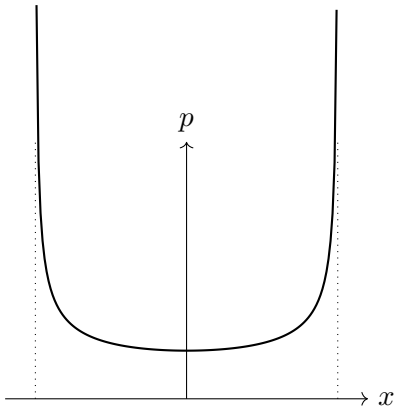
Remark. These are the invariant measures with piecewise constant density found by the Z-matrix method. Singular invariant measures also exist for any map with periodic orbits (e.g. the atomic measure at the fixed point d), but they are not of this form.

Exercise 6.7

Find an invariant measure for $f(x) = 2 - x^2$ on $[-2,2]$.



$f(x) = 2 - x^2$



$$p(x) = \frac{1}{\pi\sqrt{4-x^2}}$$

Solution.

Answer. The probability measure with density

$$p(x) = \frac{1}{\pi\sqrt{4-x^2}} \quad (-2 < x < 2), \qquad \mu(S) = \int_S p(x)\,\mathrm{d}x,$$

is f-invariant. ($\int_{-2}^{2} p = \frac{1}{\pi}[\arcsin\frac{x}{2}]_{-2}^{2} = 1$.)

Verification 1: transfer-operator check. f maps $[-2,2]$ into itself ($f(0) = 2$, $f(\pm 2) = -2$). For $y \in [-2,2]$ the preimages are $x = \pm\sqrt{2-y}$, both in $[-2,2]$, and $|f'(x)| = 2|x| = 2\sqrt{2-y}$. A density p is invariant iff

$$p(y) = \sum_{f(x)=y} \frac{p(x)}{|f'(x)|} = 2 \cdot \frac{p(\sqrt{2-y})}{2\sqrt{2-y}} = \frac{p(\sqrt{2-y})}{\sqrt{2-y}}.$$

With $p(\sqrt{2-y}) = \dfrac{1}{\pi\sqrt{4-(2-y)}} = \dfrac{1}{\pi\sqrt{2+y}}$ the right side is $\dfrac{1}{\pi\sqrt{2+y}\sqrt{2-y}} = \dfrac{1}{\pi\sqrt{4-y^2}} = p(y)$. ✓

Verification 2: directly on intervals, via a conjugacy. Write $x = 2\cos\theta$ with $\theta \in [0,\pi]$. Then $f(x) = 2 - 4\cos^2\theta = -2\cos 2\theta = 2\cos(\pi - 2\theta)$, i.e. $\theta \mapsto \pi - 2\theta$ reflected into $[0,\pi]$; with $t = \theta/\pi \in [0,1]$ this is the map $t \mapsto |1-2t|$, which preserves length on $[0,1]$ (each point has two preimages and slope ± 2). The measure μ is exactly the push-forward of length on $t \in [0,1]$ (since $\mathrm{d}x = -2\pi\sin(\pi t)\mathrm{d}t$, $|\mathrm{d}t| = \frac{\mathrm{d}x}{\pi\sqrt{4-x^2}}$). Concretely, for $[a,b] \subset [-2,2]$ put $a = 2\cos\alpha$, $b = 2\cos\beta$ with $0 \le \beta < \alpha \le \pi$. Then

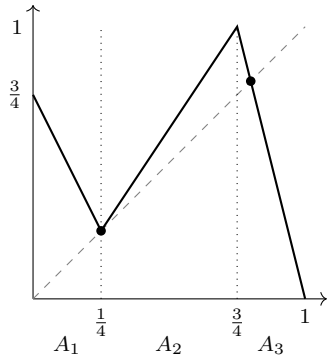
$$\mu([a,b]) = \frac{\alpha-\beta}{\pi}, \qquad f^{-1}([a,b]) = \{x : \sqrt{2-b} \le |x| \le \sqrt{2-a}\},$$

and since $\sqrt{2-2\cos\beta} = 2\sin(\beta/2)$, $\mu(f^{-1}[a,b]) = 2\cdot\frac{1}{\pi}\big(\frac{\alpha}{2} - \frac{\beta}{2}\big) = \frac{\alpha-\beta}{\pi} = \mu([a,b])$. ✓

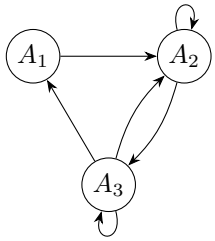
Remark. With $x = 4u - 2$ the map becomes the logistic map $u \mapsto 4u(1-u)$, and this p becomes the density $\frac{1}{\pi\sqrt{u-u^2}}$ of Challenge 6 (this is the natural measure). Trivial invariant measures, like the atomic measure at the fixed point $x = 1$, also exist.

Exercise 6.8

Let $f(x) = \frac{3}{4} - 2x$ on $[0,\frac{1}{4}]$, $\frac{3}{2}x - \frac{1}{8}$ on $[\frac{1}{4},\frac{3}{4}]$, $4 - 4x$ on $[\frac{3}{4},1]$. (a) Sketch f. (b) Find a partition and transition graph. (c) For what periods does f have a periodic orbit? (d) Find the minimum stretching factor α; is $[0,1]$ a chaotic attractor? (e) Find the natural measure.



(a) graph of f



(b) transition graph

Solution.

Values at the break points. $f(0) = \frac{3}{4}$, $f(\frac{1}{4}) = \frac{1}{4}$ (from both formulas), $f(\frac{3}{4}) = 1$ (both formulas), $f(1) = 0$. So f is continuous and its graph is the polygon $(0,\frac{3}{4}) \to (\frac{1}{4},\frac{1}{4}) \to (\frac{3}{4},1) \to (1,0)$ with slopes -2, $+\frac{3}{2}$, -4. The point $\frac{1}{4}$ is a fixed point and $\{0,\frac{3}{4},1\}$ is a period-3 orbit $(0 \to \frac{3}{4} \to 1 \to 0)$.

(a) See the graph above.

(b) Partition and transition graph. Take $A_1 = [0,\frac{1}{4}]$, $A_2 = [\frac{1}{4},\frac{3}{4}]$, $A_3 = [\frac{3}{4},1]$ (the break points are exactly $\{0,\frac{1}{4},\frac{3}{4},1\}$). Then

$$f(A_1) = [\tfrac{1}{4},\tfrac{3}{4}] = A_2, \qquad f(A_2) = [\tfrac{1}{4},1] = A_2 \cup A_3, \qquad f(A_3) = [0,1] = A_1 \cup A_2 \cup A_3.$$

So $\{0,\frac{1}{4},\frac{3}{4},1\}$ is a stretching partition and the transition graph is $A_1 \to A_2$; $A_2 \to A_2, A_3$; $A_3 \to A_1, A_2, A_3$.

(c) Periods. *Every period $n \ge 1$ occurs.* The first few periods can be written down explicitly:

period	periodic orbit(s)
1	$x = \frac{1}{4}$ and $x = \frac{4}{5}$ (from $4 - 4x = x$)
2	$\{\frac{9}{14}, \frac{47}{56}\} \approx \{0.6429, 0.8393\}$ (loop $A_2 \to A_3 \to A_2$: $\frac{9}{2} - 6x = x$)
3	$\{0, \frac{3}{4}, 1\}$ ($0 \to \frac{3}{4} \to 1 \to 0$)

For $n \geq 4$ use the closed path $A_2 \to A_3 \to A_3 \to \cdots \to A_3 \to A_2$ of length n (allowed: $A_2 \to A_3$, $A_3 \to A_3$, $A_3 \to A_2$), with word $w_n = A_2 A_3^{n-1}$, which is primitive because it contains exactly one A_2. Lemma A gives x with $f^n(x) = x$ and $f(x), \ldots, f^{n-1}(x) \in A_3$. We must show the orbit avoids the partition points $\{0, \frac{1}{4}, \frac{3}{4}, 1\}$ in order to apply Lemma B. The only periodic ones are the fixed point $\frac{1}{4}$ and the 3-cycle $\{0, \frac{3}{4}, 1\}$. The orbit of $\frac{1}{4}$ stays at $\frac{1}{4} \notin A_3$, so it cannot be our orbit (we need $f(x) \in A_3$). Any three consecutive points of the 3-cycle include $0 \notin A_3$, whereas $n - 1 \geq 3$ consecutive iterates of x lie in A_3. Hence the orbit of x avoids all partition points, and Lemma B gives minimal period exactly n.

Answer: periodic orbits of every period $n = 1, 2, 3, \ldots$. (A computer search of all linear branches of f^n for $n \leq 9$ confirms it.)

(d) Stretching factor and chaotic attractor. The slopes are -2, $\frac{3}{2}$, -4 in absolute value, so the minimum stretching factor is $\boxed{\alpha = \frac{3}{2}} > 1$. The transition graph contains the 3-cycle $A_1 \to A_2 \to A_3 \to A_1$, which passes through every symbol, so any two symbols (including a symbol with itself) can be connected both ways. By Theorem 6.11(3), f has a dense chaotic orbit and $I = [0, 1]$ is a **chaotic attractor** (it attracts itself, a set of positive length).

(e) Natural measure. The lengths are $L = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ and the slopes $s = (2, \frac{3}{2}, 4)$. The Z-matrix (column j has $1/s_j$ in each row i with $A_j \to A_i$) is

$$Z = \begin{pmatrix} 0 & 0 & \frac{1}{4} \\ \frac{1}{2} & \frac{2}{3} & \frac{1}{4} \\ 0 & \frac{2}{3} & \frac{1}{4} \end{pmatrix}.$$

(It is not column-stochastic, but DZD^{-1} with $D = \operatorname{diag}(L)$ is: e.g. column 2 of DZD^{-1} sums to $\frac{L_2 + L_3}{s_2 L_2} = \frac{3/4}{3/4} = 1$.) Solve $Zp = p$, i.e. equations (Z):

$$p_1 = \tfrac{1}{4} p_3, \qquad p_2 = \tfrac{1}{2} p_1 + \tfrac{2}{3} p_2 + \tfrac{1}{4} p_3, \qquad p_3 = \tfrac{2}{3} p_2 + \tfrac{1}{4} p_3.$$

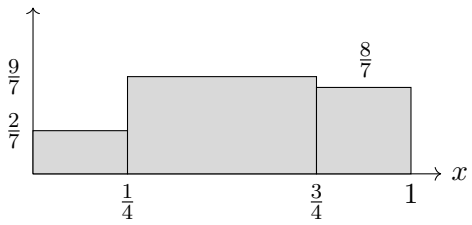
The third gives $p_3 = \frac{8}{9} p_2$, the first $p_1 = \frac{2}{9} p_2$, and the second then holds identically ($\frac{1}{9} + \frac{2}{3} + \frac{2}{9} = 1$). So $p \propto (2, 9, 8)$. Normalize: $\sum p_i L_i = (2 \cdot \frac{1}{4} + 9 \cdot \frac{1}{2} + 8 \cdot \frac{1}{4}) k = 7k = 1$, so $k = \frac{1}{7}$:

$$\boxed{p(x) = \begin{cases} \frac{2}{7}, & 0 \leq x \leq \frac{1}{4} \\ \frac{9}{7}, & \frac{1}{4} \leq x \leq \frac{3}{4} \\ \frac{8}{7}, & \frac{3}{4} \leq x \leq 1 \end{cases}}$$

The other eigenvalues: $\operatorname{tr} Z = \frac{11}{12}$ and $\det Z = \frac{1}{12}$ give, besides 1, the roots of $\lambda^2 + \frac{1}{12}\lambda + \frac{1}{12} = 0$, i.e. $\lambda = \frac{-1 \pm i\sqrt{47}}{24}$ with $|\lambda| = \frac{1}{\sqrt{12}} \approx 0.289 < 1$. So 1 is a simple eigenvalue and all others are smaller in modulus: **this is the natural measure of f.**

Check of invariance. Take $S = [\frac{1}{4}, \frac{1}{2}]$: $\mu(S) = \frac{9}{7} \cdot \frac{1}{4} = \frac{9}{28}$. Its preimage is $[\frac{1}{8}, \frac{1}{4}] \subset A_1$ (from $\frac{3}{4} - 2x \in S$), $[\frac{1}{4}, \frac{5}{12}] \subset A_2$ (from $\frac{3}{2}x - \frac{1}{8} \in S$) and $[\frac{7}{8}, \frac{15}{16}] \subset A_3$ (from $4 - 4x \in S$), with measures

$\frac{2}{7}\cdot\frac{1}{8}+\frac{9}{7}\cdot\frac{1}{6}+\frac{8}{7}\cdot\frac{1}{16}=\frac{1}{28}+\frac{6}{28}+\frac{2}{28}=\frac{9}{28}$. ✓As a by-product, the Lyapunov exponent of f is $\int \ln|f'|\,\mathrm{d}\mu=\frac{1}{14}\ln 2+\frac{9}{14}\ln\frac{3}{2}+\frac{2}{7}\ln 4\approx 0.706>0$.



density $p(x)$ of the natural measure of Exercise 6.8

Exercise 6.9

Let $p\in\mathbb{R}^n$. Prove that assigning 0 to each set not containing p and 1 to each set containing p is a probability measure on $\mathbb{R}^n$ (the atomic measure at p).

Solution. Define, for $S\subset\mathbb{R}^n$ (in particular for every closed set, and every set obtainable from them by countable unions/complements),

$$\mu(S)=\begin{cases}1, & p\in S\\ 0, & p\notin S.\end{cases}$$

We verify the defining properties.

(a) Nonnegativity. $\mu(S)\in\{0,1\}$, so $\mu(S)\geq 0$.

(b) Additivity. Let $S_1,S_2,\dots$ be a finite or countable family of pairwise disjoint sets. Because they are disjoint, the point p belongs to *at most one* of them.

- If $p\in\bigcup_i S_i$, then $p\in S_k$ for exactly one k. Then $\mu(\bigcup_i S_i)=1$, and $\sum_i\mu(S_i)=\mu(S_k)+\sum_{i\neq k}0=1$.
- If $p\notin\bigcup_i S_i$, then $p\notin S_i$ for every i, so $\mu(\bigcup S_i)=0=\sum_i\mu(S_i)$.

In both cases $\mu(\bigcup S_i)=\sum\mu(S_i)$. (Also $\mu(\emptyset)=0$, taking the empty family.)

(c) Total mass. $p\in\mathbb{R}^n$, so $\mu(\mathbb{R}^n)=1$.

Hence μ satisfies (a), (b), (c), and is a probability measure; it is *atomic* because all of its mass sits on the single point p. □

Remarks. (1) By Exercise T6.8, any set not containing p is μ-null, any set containing p has μ-measure 1—this is exactly the behaviour found in Exercises 6.4 and 6.5, where f has a natural measure that is the atomic measure at the sink. (2) μ is f-invariant (property (d)) iff $f(p)=p$: $\mu(f^{-1}(S))=1\iff f(p)\in S$, which agrees with $\mu(S)=1\iff p\in S$ for all S exactly when $f(p)=p$.

Numerical sanity checks (not part of the proofs)

I confirmed the following by computer: the sets of minimal periods in 6.2 and 6.8 (by solving $f^n(x)=x$ on every linear branch for $n\leq 9$ or 10); the formulas for $g=f^2$ in 6.6 (agreement

to 10^{-15}); the transition-matrix eigenvalues and the densities in 6.8 (histogram of f^{25} of 4×10^6 random points: $(0.285, 0.285, 1.286, \dots)$ against the predicted $\frac{2}{7}, \frac{9}{7}, \frac{8}{7}$); the bin masses of the density in 6.7 (largest discrepancy 2×10^{-4}); and the convergence statements in 6.3 and 6.5.